

Sample Solutions for Homework #1

Question 1. Write BNF or EBNF description for “struct” definition in C language.

Answer:

```
<struct> -> struct <name> <start> <var_definiton> <end>
<start> -> {
<end> -> }
<name> -> <symbol> | <symbol> <rest>
<symbol> -> _ | <lowercase> | <uppercase>
<lowercase> -> a | b | ... | z
<uppercase> -> A | B | ... | Z
<rest> -> <uppercase> | <lowercase> | <symbol> |
        <uppercase> <rest> |
        <lowercase> <rest> |
        <symbol> <rest>

<var_definition> -> <type> <name> | <type> <name> <var_definition>
<type> -> int | float | double | char
```

Question 2. Describe, in English, the language defined by the following grammar:

$\langle S \rangle \rightarrow \langle A \rangle \langle S \rangle \langle B \rangle \mid \epsilon$

$\langle A \rangle \rightarrow a$

$\langle B \rangle \rightarrow b$

For the above grammar ϵ means null.

Answer:

By using the grammar given in this question, you can generate strings like $a^n b^n$ for $n > 0$ which means that, the string starts with a, followed by any number of a's, then followed by any number of b's such that the number of a's and b's are equal to each other. Empty string (or null) is also accepted by the grammar.

Question 3. Prove that the following grammar is ambiguous:

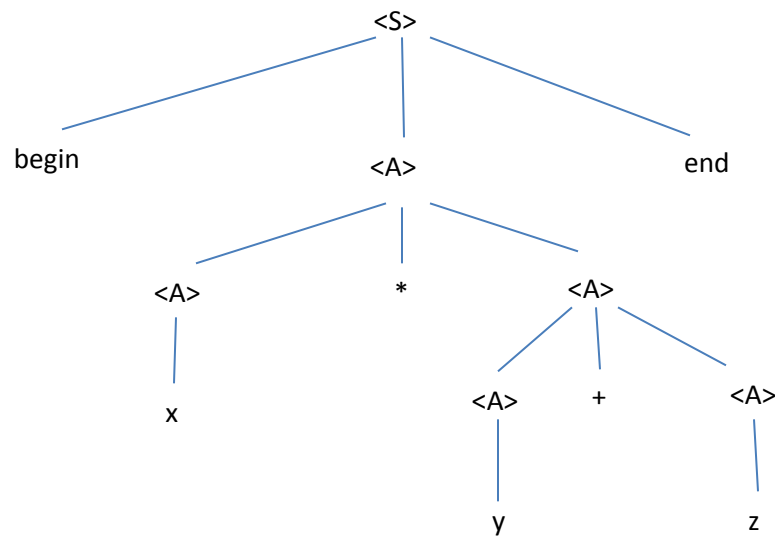
$\langle S \rangle \rightarrow \text{begin } \langle A \rangle \text{ end}$

$\langle A \rangle \rightarrow \langle A \rangle * \langle A \rangle \mid \langle A \rangle + \langle A \rangle \mid \langle A \rangle - \langle A \rangle \mid \langle A \rangle / \langle A \rangle \mid \langle id \rangle$

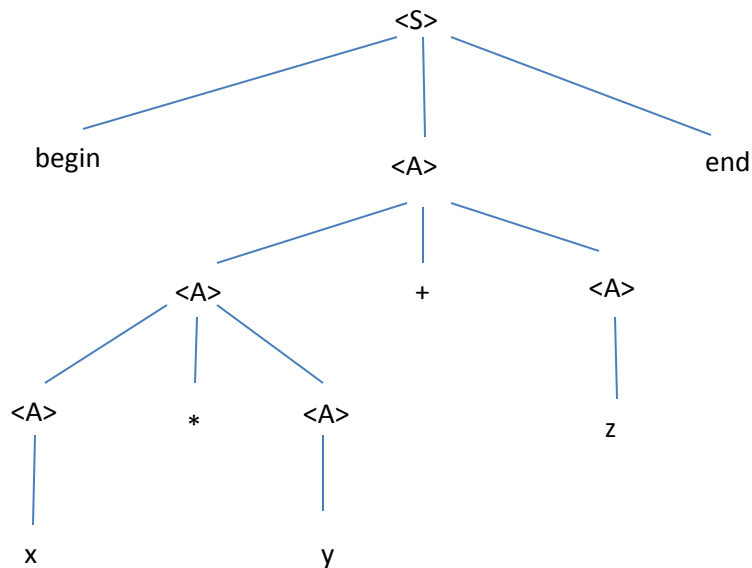
$\langle id \rangle \rightarrow x \mid y \mid z$

Answer:

If we can find a string that is generated by the given grammar such that the string has two different parse trees (according to the given grammar), then we prove that the grammar is ambiguous. Let the string be **begin x * y + z end** which is generated by the given grammar. The two different parse trees for the given string are as follows:



According to the above parse tree, first $y + z$ is computed, after that the result is multiplied with x . The second parse tree is shown below:



In the above parse tree, $x * y$ is computed first, then the result is summed with z . Since we have two different computations for the same string, the grammar is ambiguous.

Question 4. Compute the weakest precondition for the following if – else statement:

if (a == b)

b = 3 * a + 2;

else

b = 3 * a;

{b > 3}

Answer:

If (a == b) then $b = 3 * a + 2$ is computed and the post condition becomes $\{ b > 3 \}$ so,

$$3 * a + 2 > 3 \quad \text{therefore} \quad \{a > \frac{1}{3}\}$$

Otherwise, $b = 3 * a$ is computed and the post condition becomes $\{ b > 3 \}$ so,

$$3 * a > 3 \quad \text{so,} \quad \{a > 1\}$$

Therefore $\{ a > 1 \}$ is the weakest precondition